

Opportunities in PL+networking

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matters to people's real lives

systems

systems (e.g., Nuance, now part of MSFT)

goods, and services around the world

s count on you do

Quality is job #1

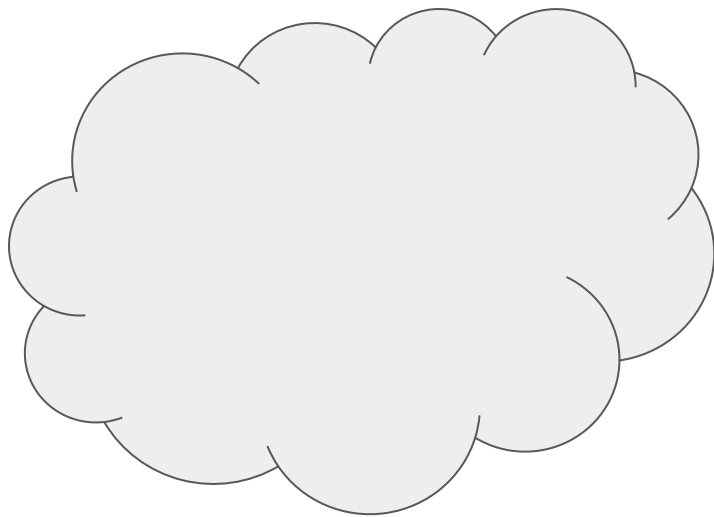
Our work here matters to people's real lives

- 911 dispatch systems
- Hospital and medical IT systems
- Doctor's office dictation systems (e.g., Nuance, now part of MSFT)
- The movement of people, goods, and services around the world
- Major retail
- Banking and fintech

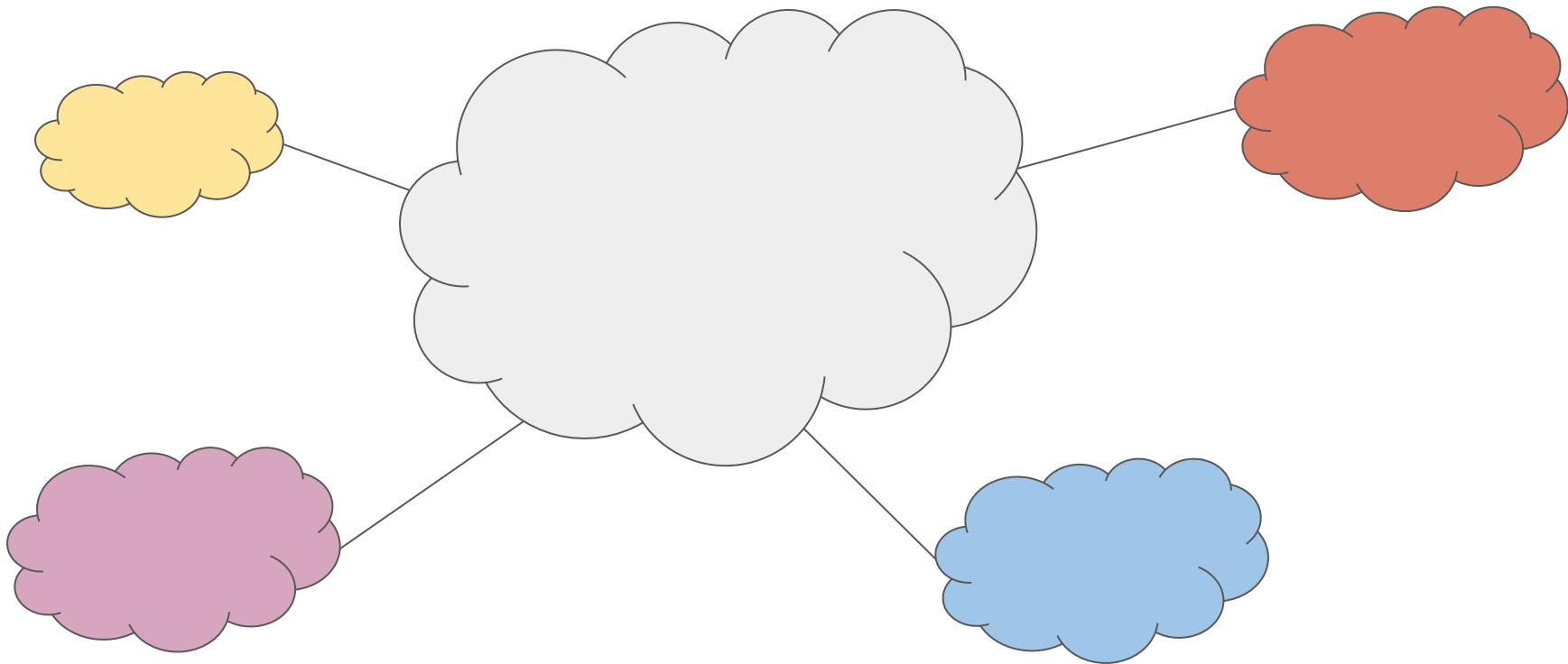
Your neighbors, your families count on you doing quality work



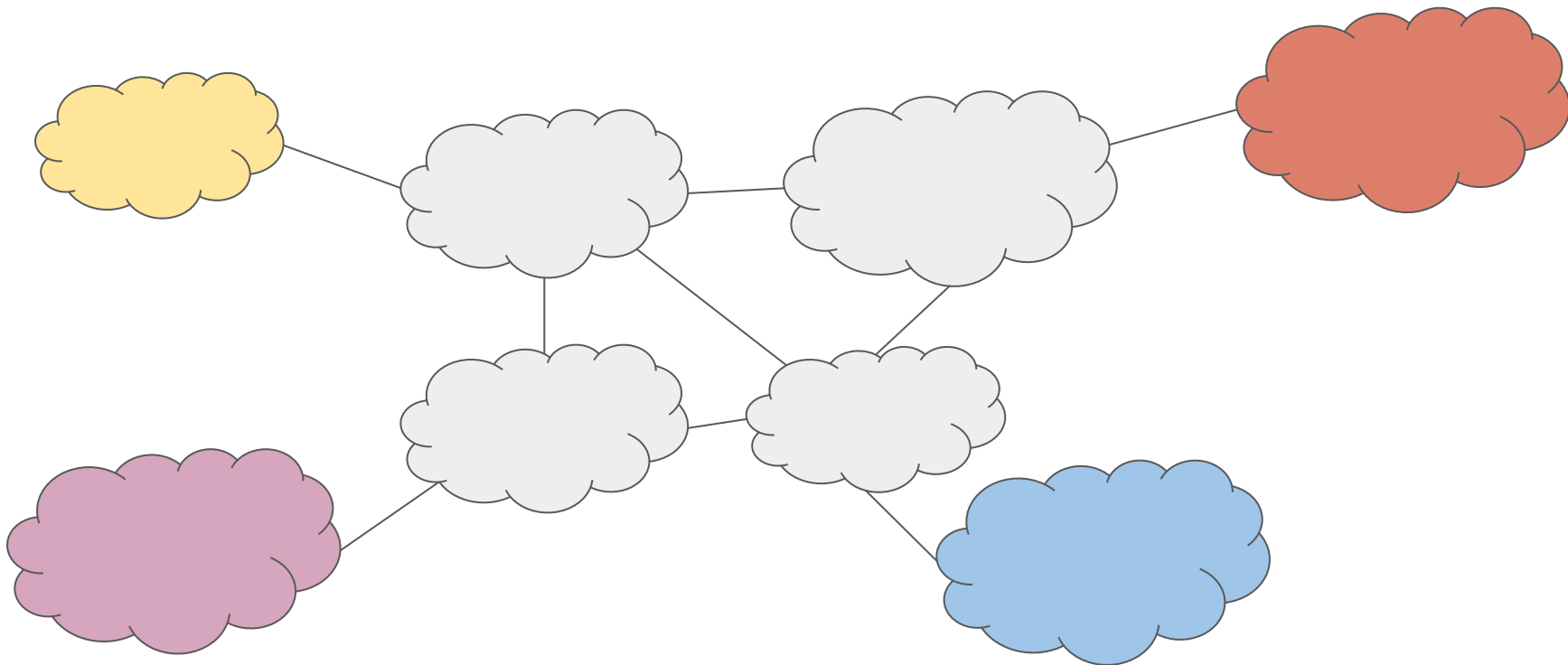
Modular reasoning



Modular reasoning



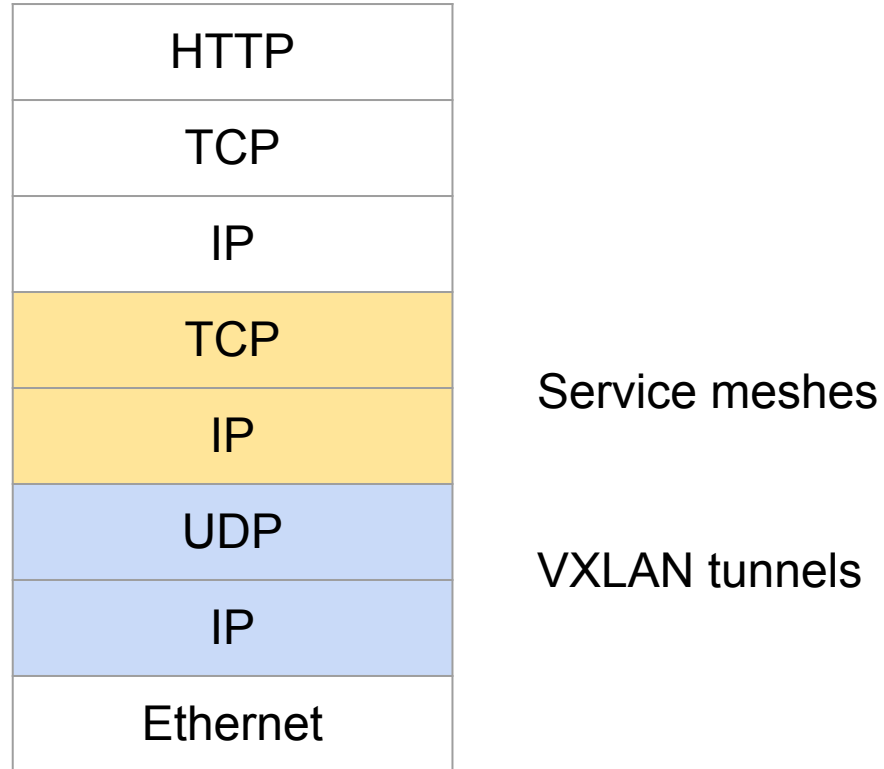
Modular reasoning



Modular reasoning

HTTP
TCP
IP
Ethernet

Modular reasoning



Modular reasoning: Advantages

Scalability

End-to-end properties

Local checking and enforcement

Debuggability

Modular reasoning: Research agenda

Abstractions for “domains”

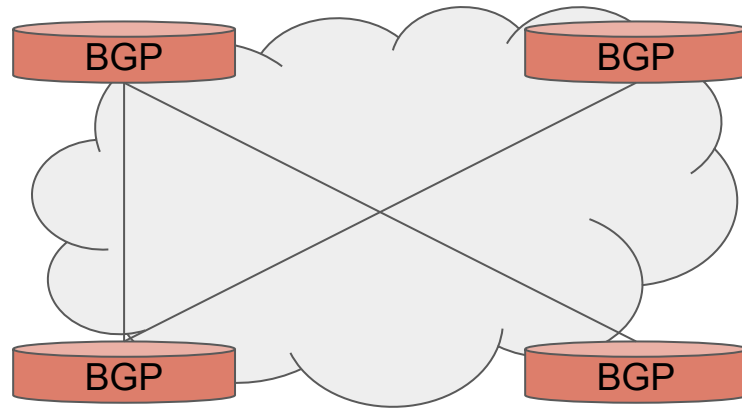
Formal interfaces between domains

Synthesis and verification

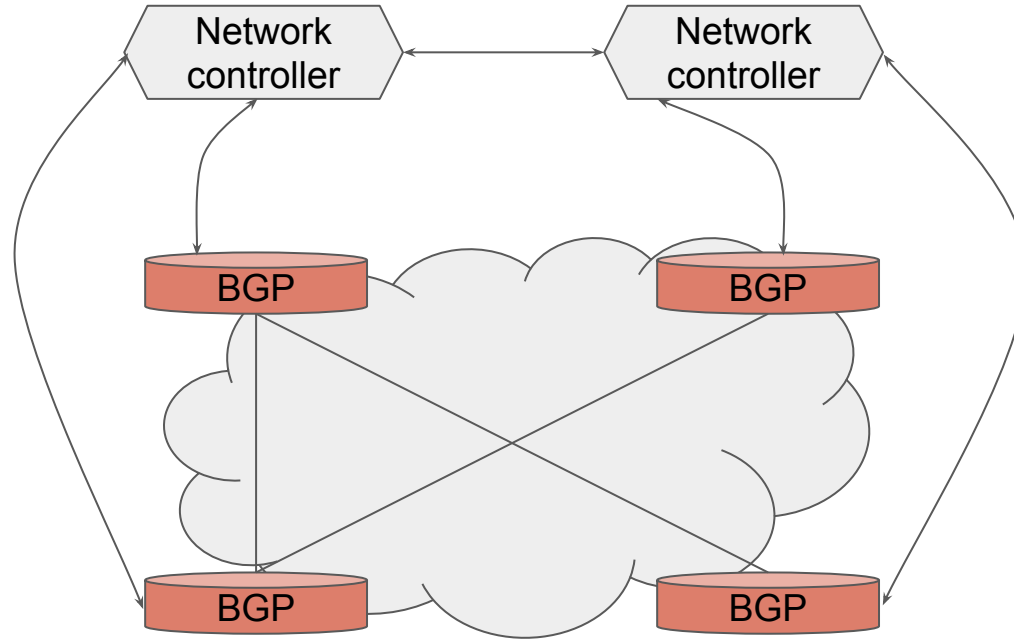
Handling uncertainty and incomplete information

New functionality

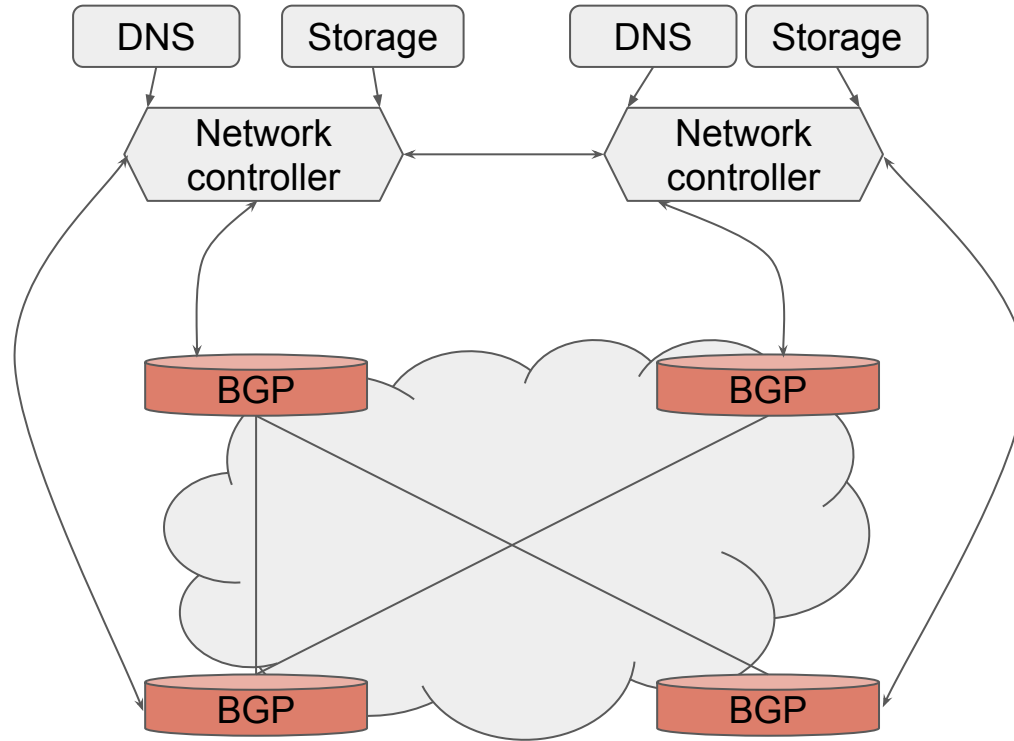
Modern control planes



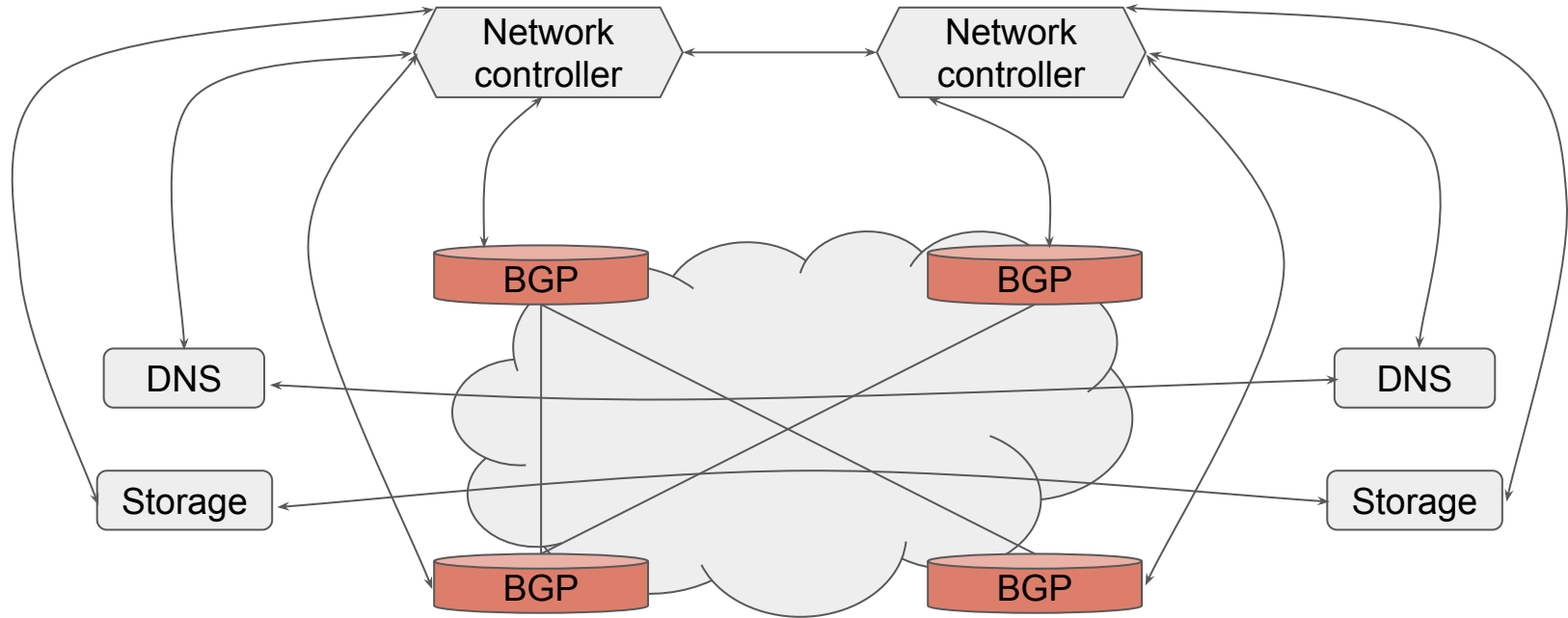
Modern control planes



Modern control planes



Modern control planes



Facebook outage: what went wrong and why did it take so long to fix after social platform went down?

Billions of users were unable to access Facebook, Instagram and WhatsApp for hours while the social media giant scrambled to restore services



Facebook, Instagram and WhatsApp all went down, and reappeared online after a six-hour global outage. Photograph: Anadolu Agency/Getty Images

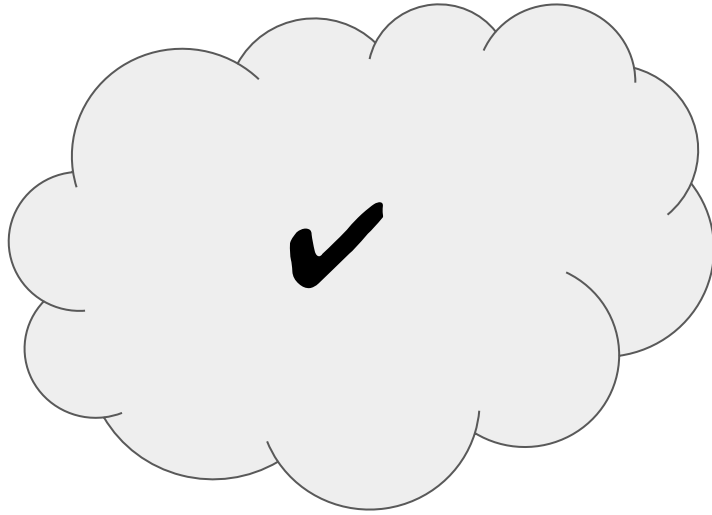
Modern control planes: Research agenda

Formalize dependencies

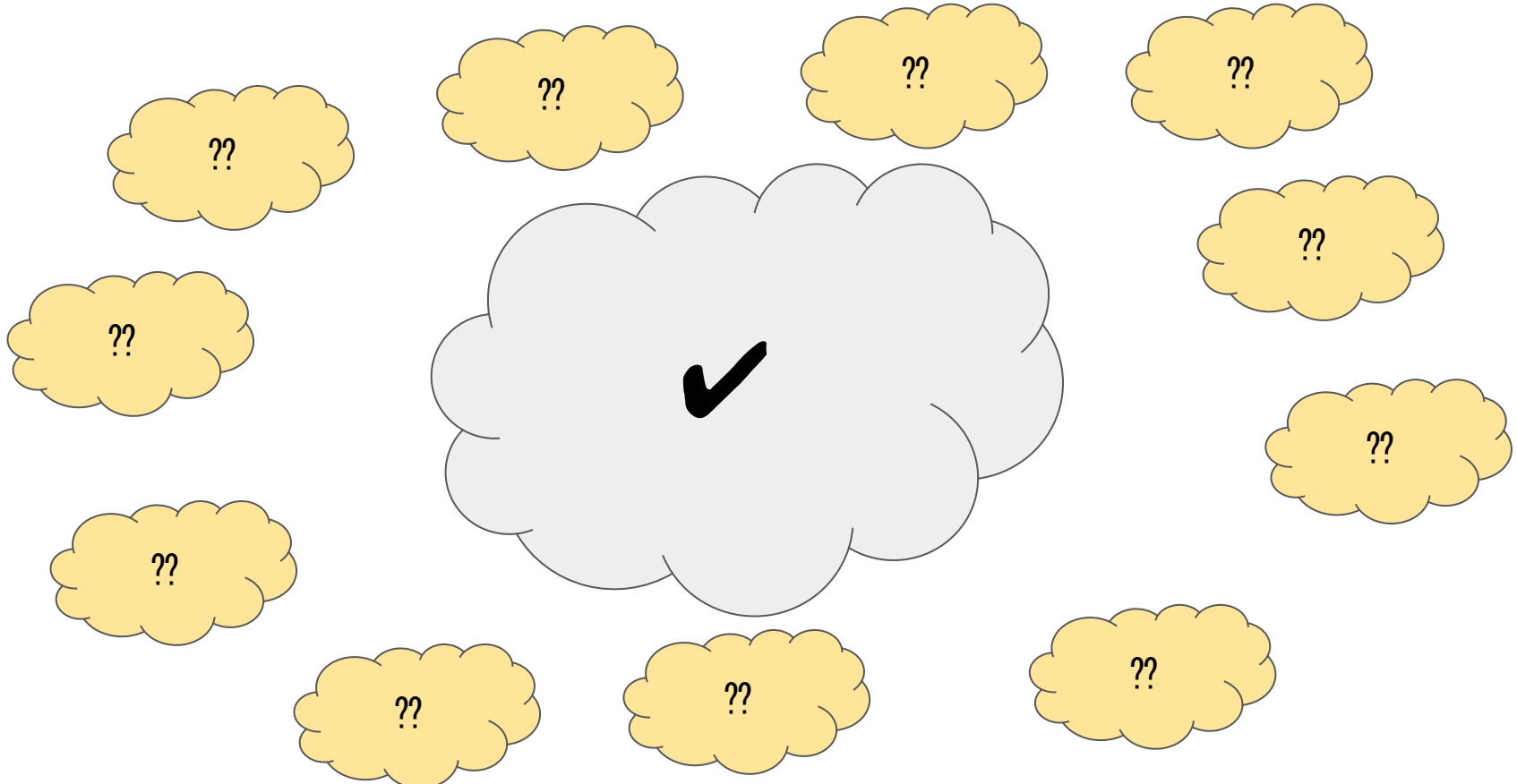
Prove resilience to all k -failure combos and to disasters

Static stability

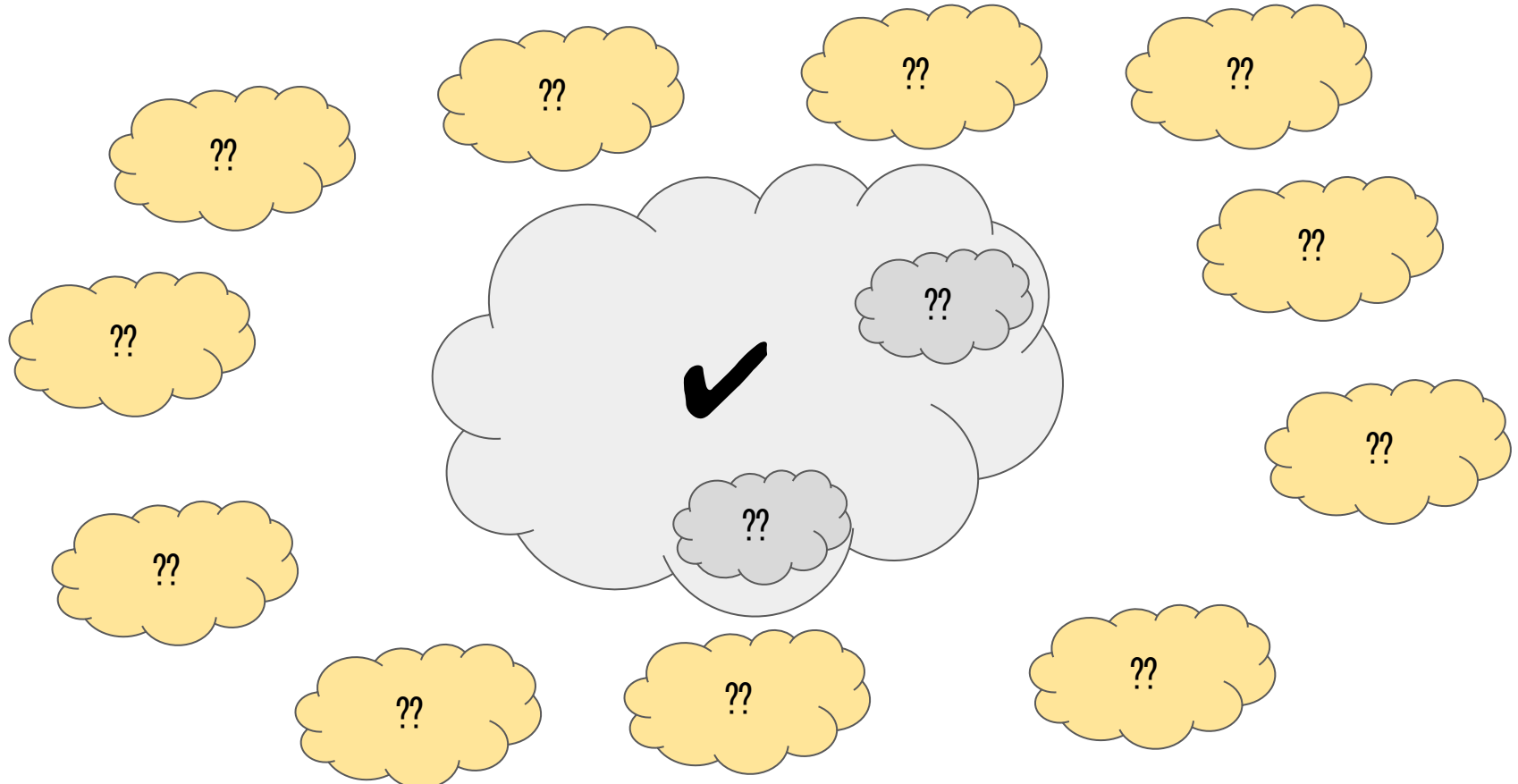
Network verification beyond the hyperscalers



Network verification beyond the hyperscalers



Accessible network verification



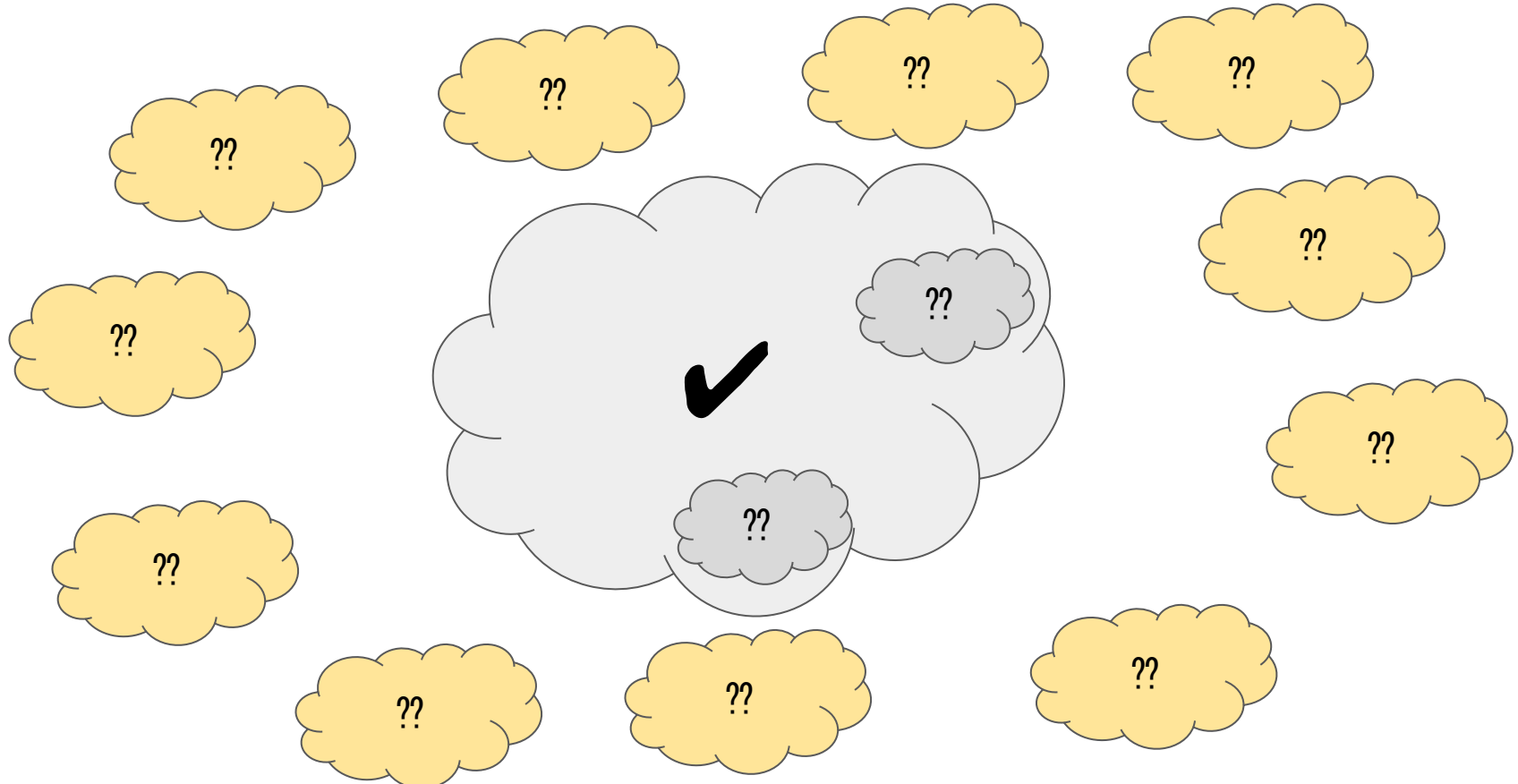
Network verification beyond the hyperscalers: Challenges

Lack of precise specifications

Low user expertise

New mental models and abstractions

Network verification beyond the hyperscalers



Network verification beyond the hyperscalers: Agenda

Inspiration from software testing

- Type systems
- Testing frameworks
- Programming environments

Trade strong guarantees for accessibility

Verification without formal specifications

PL+networking+*HCI*

Ecosystem of languages for networks

Well-designed languages facilitate

- Reuse, composition, fast and safe refactoring
- “Correct by construction” design

Examples

- Switch data plane → P4, ...
- Network-wide data plane → NetKAT,
- Network-wide BGP → Propane

Ecosystem of languages for networks

Many unexplored domains

- Middleboxes
- Service meshes (application-layer networks)
- Richer control planes
- “Enhanced Internet”

Beyond individual languages: A thriving ecosystem of languages

- “There is a DSL for that”
- Shared intermediate representation(s)
- Programming environments

Summary

Modular reasoning

Verification for modern control planes

Network verification beyond hyperscalers

Thriving ecosystem of languages for networks